

Subsets of Real Numbers

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The set of **Real Numbers** ($\mathbb{R}$) consists of all numbers that can be located on the number line.

Natural Numbers

The set of **Natural Numbers** ($\mathbb{N}$) is

$$\mathbb{N} = \{1, 2, 3, \dots\}.$$

Whole Numbers

The set of **Whole Numbers** ($\mathbb{W}$) is

$$\mathbb{W} = \{0, 1, 2, 3, \dots\}.$$

Integers

The set of **Integers** ($\mathbb{Z}$) includes negative whole numbers, zero, and positive whole numbers.

$$\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}.$$

Rational Numbers

The set of **Rational Numbers** ($\mathbb{Q}$) consists of numbers that can be written as

$$\frac{a}{b}$$

where a and b are integers and $b \neq 0$.

For example,

$$\frac{3}{4}, \quad -\frac{5}{2}, \quad 7.$$

Irrational Numbers

The set of **Irrational Numbers** consists of numbers that cannot be written as a ratio of two integers.

Their decimal forms are non-terminating and non-repeating.

Examples include

$$\sqrt{2}, \quad \sqrt{5}, \quad \pi.$$

Subsets of Real Numbers

Every natural number is also a whole number.

Every whole number is an integer.

Every integer is a rational number.

Every rational number is a real number.

Nested Diagram

The subsets of real numbers can be illustrated using a nested diagram:

$$\mathbb{N} \subset \mathbb{W} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}.$$

Thus, every natural number is a real number.

Examples

Example 1

Classify the number

9.

Since 9 is a natural number, it is also whole, integer, rational, and real.

The smallest set listed is

Natural.

Example 2

Classify the number

-6 .

The number -6 is an integer.

Therefore, it is also rational and real.

The smallest set listed is

Integer.

Example 3

Classify the number

$$\frac{5}{8}.$$

It can be written as a ratio of two integers.

Therefore,

$$\frac{5}{8}$$

is a **rational number**.

Example 4

Classify the number

$$\sqrt{2}.$$

It cannot be written as a ratio of two integers.
Therefore,

$$\sqrt{2}$$

is an **irrational number**.

Practice Exercises

Practice 1

Classify each number as **Natural**, **Whole**, **Integer**, **Rational**, or **Irrational**.

(a) 9 (b) 0

(c) -6 (d) $\frac{5}{8}$

(e) $\sqrt{2}$

Practice 2

Write all the subsets of real numbers to which each number belongs.

(a) 12 (b) 0

(c) -9 (d) $\frac{7}{4}$

(e) π

Practice 3

Identify whether each statement is **True** or **False**.

- (a) Every whole number is an integer.
- (b) Every rational number is an integer.
- (c) Every irrational number is a real number.
- (d) Zero is a natural number.
- (e) Every integer is rational.

Practice 4

Classify each decimal number.

(a) 0.75 (b) -3.2

(c) $0.121212\dots$

(d) $1.414213\dots$

(e) 8.0

Practice 5

Which set best describes each number?

(a) $\sqrt{36}$ (b) $-\sqrt{49}$

(c) $\frac{-10}{5}$ (d) $\sqrt{5}$

(e) $\frac{11}{2}$

Practice 6

Which numbers belong to the set of integers?

$-8,$ $0,$ $\frac{2}{3},$ $15,$ $\sqrt{7},$ -21

Practice 7

Which numbers are irrational?

$$\sqrt{11}, \quad \frac{9}{10}, \quad \pi,$$

$$4, \quad \sqrt{49}, \quad \sqrt{13}$$

Activities

Activity 1

Classify each number as **Natural**, **Whole**, **Integer**, **Rational**, or **Irrational**.

(a) 14 (b) 0

(c) -15 (d) $\frac{9}{11}$

(e) $\sqrt{3}$

Activity 2

Write all the subsets of real numbers to which each number belongs.

(a) 20 (b) 0

(c) -18 (d) $\frac{13}{5}$

(e) $\sqrt{17}$

Activity 3

Identify whether each statement is **True** or **False**.

(a) Every natural number is a whole number.

(b) Every irrational number is rational.

(c) Every integer is a real number.

(d) Zero is a whole number.

(e) Every rational number is real.

Activity 4

Classify each decimal number.

(a) 0.625 (b) -7.4

(c) $0.454545\dots$

(d) $3.141592\dots$

(e) 11.0

Activity 5

Which set best describes each number?

(a) $\sqrt{81}$ (b) $-\sqrt{64}$

(c) $-\frac{24}{6}$ (d) $\sqrt{19}$

(e) $\frac{17}{3}$

Activity 6

Which numbers belong to the set of integers?

-30 , 8 , $\frac{5}{6}$, 0 , $\sqrt{15}$, 27

Activity 7

Which numbers are irrational?

$$\sqrt{10}, \quad \frac{4}{7}, \quad \pi,$$

$$\sqrt{64}, \quad \sqrt{21}, \quad 9$$

Thank you!